

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2024-P1-Model)** **Date: 29-04-2026**

Time: 09.00Am to 12.00Pm **GTA-28** **Max. Marks: 180**

29-04-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2024-P1)_GTA-28_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

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Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2024-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Matching Type	+3	-1	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Matching Type	+3	-1	4	12
				17	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Matching Type	+3	-1	4	12
Total				**17**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. The area of the figure in the complex plane enclosed by the origin and the set of all points $\frac{1}{z}$ such that $(1-2i)z+(-2i-1)\bar{z}=6i$, is

A) $\frac{\pi}{36}$ **B)** $\frac{5\pi}{36}$ **C)** $\frac{5\pi}{6}$ **D)** $\frac{\pi}{6}$

2. Three students S_1, S_2 and S_3 are given a problem to solve. Consider the following events:

U : At least one of S_1, S_2 and S_3 can solve the problem,

V: S_1 can solve the problem, given that neither S_2 nor S_3 can solve the problem,

W: S_2 can solve the problem and S_3 cannot solve the problem,

T: S_3 can solve the problem.

For any event E, let P(E) denote the probability of E.

If $P(U)=\frac{1}{2}, P(V)=\frac{1}{10}$ and $P(W)=\frac{1}{12}$,

then $4P(\bar{T})$ is equal to

A) $\frac{13}{12}$ **B) 1** **C)** $\frac{19}{15}$ **D)** $\frac{23}{9}$

3. Let R denote the set of all real numbers. Define the function $f:\mathbb{R}\to\mathbb{R}$ by f(x) =

$$\begin{cases} 2-2x^2-x^2\sin\frac{1}{x} & \text{if } x\neq 0 \\ 2 & \text{if } x=0 \end{cases}$$

then which one of the following statements is TRUE?

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A) The function f is NOT differentiable at x = 0

B) There exist a positive real number δ , such that f is a decreasing function on the interval (0, δ)

C) For any positive real number δ , the function f is an increasing function on the interval (-δ , 0)

D) x = 0 is a point of local maxima of f

4. Let $g(x)=e^{f(x)}$ and $f(x+1)=x+f(x)\forall x\in R$. If $n\in N$, then $\frac{g'\left(n+\frac{1}{2}\right)}{g\left(n+\frac{1}{2}\right)}-\frac{g'\left(\frac{1}{2}\right)}{g\left(\frac{1}{2}\right)}$ is always equal to

A) $2\left(1+\frac{1}{2}+\frac{1}{3}+\ldots+\frac{1}{n}\right)$ **B)** $2\left(1+\frac{1}{3}+\frac{1}{5}+\ldots+\frac{1}{2n-1}\right)$

C) n **D)** 1

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. If (a, b) is a point on the circle centered at (5, 0) with radius 4 in the xy-plane, the maximum possible value of $\frac{a^2+7b^2}{2a^2+b^2}$ is $\frac{p}{q}; p,q\in N$, gcd (p, q) = 1, then which of the following is/are CORRECT?

A) p – 3q = 91 **B)** p + q = 155 **C)** 2p – 3q = 140 **D)** q = 54

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6. 5 integers are each selected uniformly at random from range 1 to 5 inclusive and put into a set S. Each integer is selected independently of the others. Which of the following is/are TRUE

A) The probability that value of minimum element is 2, is $\frac{2101}{6^5}$

B) The probability that value of minimum element is 3, is $\frac{3^5-2^5}{5^5}$

C) The expected value of minimum element of S is $\frac{177}{125}$

D) The expected value of minimum element of S is $\frac{125}{177}$

7. Let three vectors $\vec{a},\vec{b}$ and $\vec{c}$ be such that $\vec{a}\times\vec{b}=\vec{c},\vec{b}\times\vec{c}=\vec{a}$ and $|\vec{a}|=2$ then which one of the following is/are true?

A) $\vec{a}\times\left(\left(\vec{b}+\vec{c}\right)\times\left(\vec{b}-\vec{c}\right)\right)=\vec{0}$ **B)** Projection of $\vec{a}$ on $\left(\vec{b}\times\vec{c}\right)$ is 2

C) $\left[\vec{a}\vec{b}\vec{c}\right]+\left[\vec{c}\vec{a}\vec{b}\right]=8$ **D)** $\left|3\vec{a}+\vec{b}-2\vec{c}\right|^2=51$

SECTION-III
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

8. Let $(a_1,a_2,a_3.....)$ be a sequence denoted by A and a sequence ΔA is defined such that ΔA is the following sequence: $(a_2-a_1,a_3-a_2,a_4-a_3.....)$. Suppose that all of the terms of the sequence $\Delta(\Delta A)$ are 1 and it is given that $a_{19}=a_{92}=0$. Find a_3

9. For integers n and r, let

$$\binom{n}{r}=\begin{cases}{}^nC_r, & \text{if } n\geq r\geq 0\\ 0, & \text{otherwise}\end{cases}$$

The maximum value of k for which the sum

$$\sum_{i=0}^{k}\binom{10}{i}\binom{15}{k-i}+\sum_{i=0}^{k+1}\binom{12}{i}\binom{13}{k+1-i}$$ exists and not equal to 0, is equal to _____.

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10. Consider the system of equations ax + y = b, bx + y = a, ax + by = ab where $a,b\in\{0,1,2,3,4\}$. Find the number of ordered pairs (a, b) for which system is consistent.

11. $f(x)=\prod_{k=1}^{\infty}\left(\frac{1+2\cos\left(\frac{2x}{3^k}\right)}{3}\right)$, then number of points where

$[xf(x)]+|xf(x)|+(x-1)|x^2-3x+2|$ is not differentiable in $(0,2\pi)$ where [.] is GIF

12. Ram chooses five subsets S_1,S_2,S_3,S_4,S_5 of {1, 2, 3, 4, 5, 6, 7} such that $|S_1|=1,|S_2|=2,|S_3|=3,|S_4|=4$, and $|S_5|=5$. Moreover, for all $1\le i<j\le 5$, either $S_i\cap S_j=S_i$ or $S_i\cap S_j=\phi$ (in other words, the intersection of S_i and S_j is either S_i or the empty set). Number of ways can Ram select the sets is N, number of digits of N is –

13. 5 students, all with distinct ages, are randomly seated in a row at the movies. The probability that, from left to right, no three consecutive students are seated in increasing age order is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find m + n

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SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

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14. Match the following List – I with List – II.

List – I		List – II	
(P)	Number of integral value(s) of K such that all solutions (x, y) to the system of equation $x + 4y = 2K^2$ and x + y = K are such that x, y > 0 is	(1)	0
(Q)	Number of integers in the domain of function g(x) = $\ln(\cos^{-1} x)$ is	(2)	1
(R)	If x > 0; then value of expression $\frac{(1+\sin^{-1}x)^{2010}(1+\cos^{-1}x)^{2011}\left(1+\tan^{-1}x\right)^{2012}}{\left(1+\cos ec^{-1}x\right)^{2010}\left(1+\sec^{-1}x\right)^{2011}\left(1+\cot^{-1}x\right)^{2012}}$ is	(3)	2
(s)	If the function f : R – {-b} $\to$ R – {1} defined by $f(x)=\frac{x+a}{x+b},(a\neq b)$ is self inverse, then a + b can be	(4)	3
		(5)	4

A) $P\to 3;Q\to 2;R\to 1;S\to 1,3$

B) $P\to 2;Q\to 3;R\to 2;S\to 1,2,3,4,5$

C) $P\to 2;Q\to 3;R\to 1,2;S\to 4,5$

D) $P\to 2;Q\to 3,4;R\to 5;S\to 1$

15. Let $f(x)=\begin{cases}[x]; & x\in[-2,0]\\ |x|; & x\in[0,2]\end{cases}$; where [.] represent G.I.F. and g(x) = sec x,

$x\in R-(2n+1)\frac{\pi}{2}$. Match the following statements in List – I with their values in List– II in interval $\left(-\frac{3\pi}{2},\frac{3\pi}{2}\right)$.

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List – I		List – II	
(P)	Limit of fog exist at	(1)	-1
(Q)	Limit of gof does not exist at	(2)	π
(R)	Points of discontinuity of fog is/are	(3)	$\frac{5\pi}{6}$
(S)	Points of differentiability of fog is/are	(4)	$-\pi$
		(5)	$\frac{\pi}{3}$

A) $P \to 1,2,4; Q \to 1; R \to 2,4; S \to 1,3$ **B)** $P \to 1,2,4; Q \to 2,4; R \to 5; S \to 1$

C) $P \to 1,2; Q \to 1,5; R \to 2,5; S \to 1,3$ **D)** $P \to 2; Q \to 5; R \to 3; S \to 1$

16. Let $\mathbb{R}$ denote the set of all real numbers. For a real number x, let [x] denote the greatest integer less than or equal to x. Let n denote a natural number.

Match each entry in List-I to the correct entry in List-II and choose the correct option.

	List – I		List – II
(P)	Number of $x^o \in \mathbb{R}$ such that $\ell(x) = \sum_{k=0}^{4}\left(\sin\lvert x-k\rvert + \cos\left\lvert x-k+\frac{1}{2}\right\rvert\right)$, $x \in R$, is NOT differentiable at x^o, is	(1)	8
(Q)	The minimum value of n for which the function $f(x) = \left[\frac{10x^3 - 45x^2 + 60x + 35}{n}\right]$ is continuous on the interval [1, 2], is	(2)	9
	The minimum value of n for which g(x) = $\left(2n^2 - 13n - 15\right)\left(x^3 + 3x\right), x \in \mathbb{R}$, is an increasing function on R	(3)	5
(S)	The smallest natural number n which is greater than 5, such that x = 3 is a point of local minima of $h(x) = (x^2 - 9)^n (x^2 + 2x + 3)$, is	(4)	6
		(5)	10

A) $P \to 1; Q \to 3; R \to 2; S \to 5$ **B)** $P \to 3; Q \to 2; R \to 1; S \to 4$

C) $P \to 5; Q \to 1; R \to 4; S \to 3$ **D)** $P \to 2; Q \to 3; R \to 1; S \to 5$

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17. Let $\vec{w}=\hat{i}+\hat{j}-2\hat{k}$ and $\vec{u}$ and $\vec{v}$ be two vectors such that $\vec{u}\times\vec{v}=\vec{w}$ and $\vec{v}\times\vec{w}=\vec{u}$. Let α,β,γ and t be real numbers such that $\vec{u}=a\hat{i}+\beta\hat{j}+\gamma\hat{k}$, $-t\alpha+\beta+\gamma=0$, $\alpha-t\beta+\gamma=0$ and $\alpha+\beta-t\gamma=0$. Match each entry in List–I to the correct entry in List–II and choose the correct option.

	List – I		List – I
(P)	$\lvert\vec{v}\rvert^2$ is equal to	(1)	0
(Q)	If $\alpha=\sqrt{3}$, then γ^2 is equal to	(2)	1
(R)	If $\alpha=\sqrt{3}$, then $2(\beta+\gamma)^2$ is equal to	(3)	2
(S)	If $\alpha=\sqrt{2}$, then t + 5 is equal to	(4)	6
		(5)	7

A) $P\to2;Q\to1;R\to4;S\to5$ **B)** $P\to2;Q\to4;R\to3;S\to5$

C) $P\to2;Q\to1;R\to4;S\to3$ **D)** $P\to5;Q\to4;R\to1;S\to3$

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PHYSICS **Max Marks: 60**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

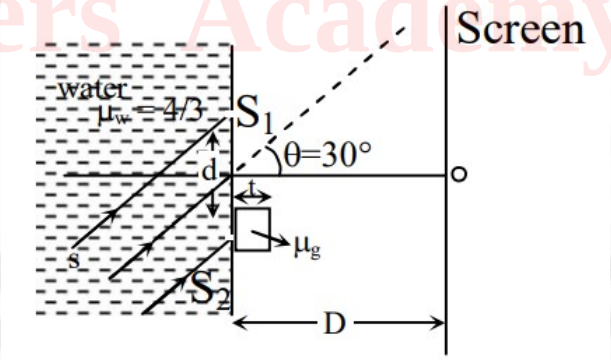
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- For each question, choose the option corresponding to the correct answer.
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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

18. In Young's experiment, the interference pattern is found to have an intensity ratio between the bright and dark fringes as 9. What is the ratio of intensities and amplitudes of the two interfering waves, respectively?

A) 4, 2 **B)** 9, 3 **C)** 2.25, 1.5 **D)** none of these

19. In a YDSE experiment, the two slits are covered with a transparent membrane of negligible thickness which allows light to pass through it but does not allow water. A glass slab of thickness t = 0.41 mm and refractive index μ_g = 1.5 is placed in front of one of the slits as shown in the figure. The separation between the slits is d = 0.30mm. The entire space to the left of the slits is filled with water of refractive index μ_w= 4/3. A coherent light of intensity I and absolute wavelength λ = 5000 A^o is being incident on the slits making an angle 30° with horizontal. If screen is placed at a distance D = 1m from the slits, then the position of central maxima will be



A) 0.58 cm above the central line of slits

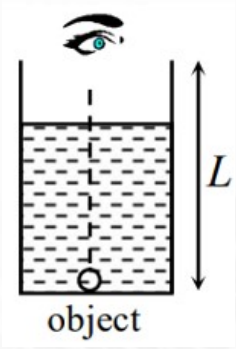
B) 0.58 cm below the central line of slits

C) 1.66 cm above the central line of slits

D) 1.66 cm below the central line of slits

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20. A cylinder is filled with a liquid of refractive index μ. The radius of the cylinder is decreasing at a constant rate K. The volume of the liquid inside the container remains constant at V. The observer and the object O are in a state of rest and at a distance L from each other. The apparent velocity of the object as seen by the observer, (when radius of cylinder is r) (considering upward direction as positive for velocity)



A) $\frac{(1-\mu)2KV}{(\pi\mu r^3)}$ **B)** $\frac{(1-\mu)2KV}{(\pi\mu Lr^2)}$ **C)** $\frac{(1-\mu)2K}{\mu}$ **D)** $\frac{(1-\mu)K}{2\mu}$

21. A car starts from rest from origin in a straight line with an acceleration (a) given by the relation $a=\frac{25}{(x+2)^3}$, where a is in m/s^2 and x is in metre. The maximum velocity of the car will be (x is the position of the car)

A) 2.5 m/s **B)** 5 m/s **C)** 10 m/s **D)** Infinite

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

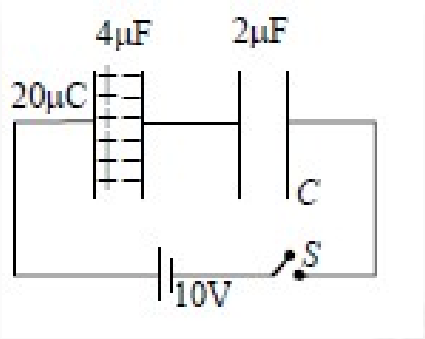
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
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For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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22. A $4\,\mu F$ capacitor is given $20\,\mu C$ charge and is connected with an uncharged capacitor of capacitance $2\,\mu F$ as shown in figure. When switch S is closed.



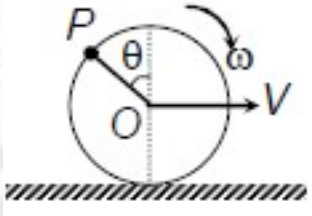
A) Charged flown through the battery is $\frac{40}{3}\mu C$

B) Charge flown through the battery is $\frac{20}{3}\mu C$

C) Work done by the battery is $\frac{200}{3}\mu J$

D) work done by the battery is $\frac{100}{3}\mu J$

23. A disc of radius R rolls on a horizontal surface with linear velocity V and angular velocity ω. There is a point P on circumference of disc at angle θ with upward vertical diameter measured anticlockwise (see figure), which has a vertical velocity. Here θ is equal to



A) $\pi + \sin^{-1}\frac{V}{R\omega}$

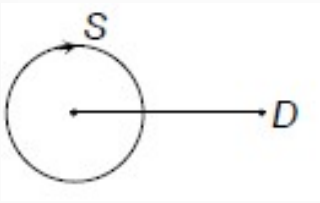
B) $\frac{\pi}{2} - \sin^{-1}\frac{V}{R\omega}$

C) $\pi - \cos^{-1}\frac{V}{R\omega}$

D) $\pi + \cos^{-1}\frac{V}{R\omega}$

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24. A source of sound moves along a circle of radius 2 m with constant angular velocity 40 rad/s. Frequency of the source is 300 Hz. A detector is kept at some distance from the circle in the same plane of the circle (as shown in figure). Which of the following is not the possible value of frequency registered by the detector? (Speed of sound = 320 m/s)



A) 250 Hz **B)** 360 Hz **C)** 410 Hz **D)** 220 Hz

SECTION-III
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

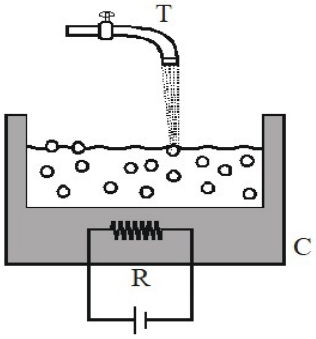
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

25. Water at temperature 40°C flows from a tap T into a heated container C. The container has a heating element (a resistor R) which is generating heat at the rate of P, that may be varied. The rate of water inflow from tap is $m = \frac{1000}{7}\ell$/min. The heat generated is sufficient so that the water in the container is boiling and getting converted into steam at a steady rate. What is the minimum power P (in MW) that must be generated as heat in the steady state in resistor R so that the amount of liquid water in the container neither increases nor decreases with time ? (Neglect other losses of heat, such as conduction from the container to the air and heat capacity of container)

For water, specific heat c = 4.2 kJ $kg^{-1}K^{-1}$,

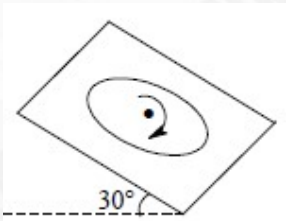
latent heat of vaporization $L_{vap} = 2.268 MJkg^{-1}$, density $\rho = 1000 kgm^{-3}$



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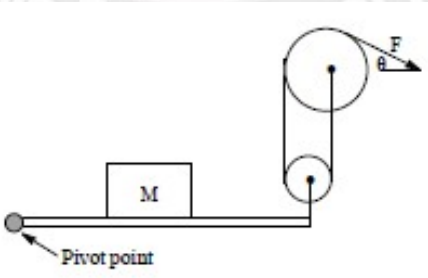
26. In a slow reaction, heat is being evolved at a rate about 10mW in a liquid. If the heat were being generated by the decay of ^{32}P, a radioactive isotope of phosphorus that has half-life of 14 days and emits only beta-particles with a mean energy of 700KeV. The number of ^{32}P atoms in the liquid is $A\times 77\times 10^{15}$ find a: [Take: ln 2 = 0.7]

27. An old record player of 10 cm radius turns at 10 rad/s while mounted on a 30° incline as shown in the figure. A particle of mass m can be placed anywhere on the rotating record. If the least possible coefficient of friction that must exist for no slipping to occur is μ, find $2\sqrt{3}\mu$ [take g = $10m/s^2$]



28. Ann is sitting on the edge of a Disc that has a radius of 6 m and is rotating steadily about its axis. Bob is standing still on the ground at a point that is 12m from the centre of the carousel. At a particular instant, Bob observes Ann moving directly towards him with a speed of $1\,ms^{-1}$. With what speed (in m/s) Bob will be moving in a rotating frame of reference in which the disc appears to be stationary?

29. A pulley system is attached to a massless board as shown below. The board pivots only at the pivot point. A 10 kg mass M sits exactly in the middle of the board. If the angle θ is 60°. The force F (in N) on the string necessary to hold the board in the position shown is $5\times x$, then find x. [take g = $10m/s^2$]



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30. The peak emission from a black body at a certain temperature occurs at a wavelength of 6000 Å. On increasing its temperature, the total radiation emitted is increased 16 times. These radiations are allowed to fall on a metal surface. Photoelectrons emitted by the peak radiation at this higher temperature can be bought to rest by applying a potential equivalent to the excitation potential corresponding to the transition for the level n = 4 to n = 2 in the Bohr's hydrogen atom. The work function of the metal is given by $\frac{\alpha}{100}eV$ where α is the numerical constant. The value of α is found $53\times x$, find x ?

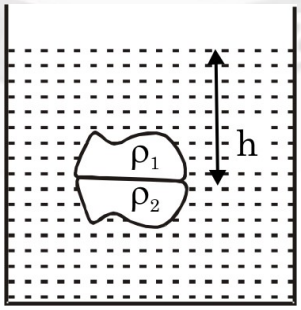
[Take: hc = 12420 eV-$\overset{o}{A}$]

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

31. Two objects of volume V_1 and V_2 and densities ρ_1 and ρ_2 have identical flat portion of area 'S' on their surfaces. These flat portions are glued to each other. The composite body thus formed, floats fully submerged in a liquid with the common flat portion horizontal as shown in figure. The glue can withstand a maximum force F_{max}. If h = minimum depth in the liquid for which the common flat portion be keeping the object intact in equilibrium and acceleration of free fall g = 10 m/s^2, column - I describe four physical quantities which have to be calculated and to be matched with column -II.



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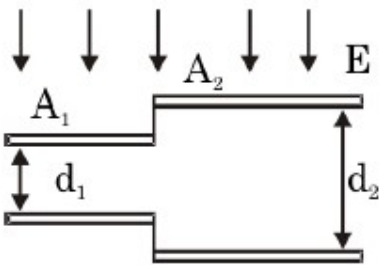
	COLUMN-I		COLUMN-II
I)	Value of h	**P)**	10 m
II)	Force by liquid on upper object	**Q)**	5500 N
III)	Force by liquid on lower object	**R)**	4500 N
IV)	Total force by liquid on both object	**S)**	10^4N
		T)	3000 N
		U)	7500 N

If object shown in figure has volume $V_1 = 1m^3$ and $V_2 = 0.5m^3$ densities $\rho_1 = 400kg/m^3$ and $\rho_2 = 700kg/m^3$ and identical flat portion has area $S = 100cm^2$, and if the glue can with stand a maximum force $F_{max} = 500$N (Neglect atmospheric pressure). Which of the following match is correct?

A) $I-P, II-T, III-U, IV-S$ **B)** $I-P, II-R, III-T, IV-U$

C) $I-P, II-R, III-Q, IV-S$ **D)** $I-P, II-Q, III-R, IV-S$

32. A parallel plate capacitor has two sections. In these sections, areas of plates are A_1 and A_2 and distances between the plates are d_1 and d_2 respectively. The distances between the plates are much smaller than their linear dimensions. A uniform electric field E that is perpendicular to the plates is switched on. Neglecting edge effects, E_1 and E_2 are electric fields established in sections (1) and (2) respectively.



List-I describes various physical quantities (in SI unit). Match it with List-II.

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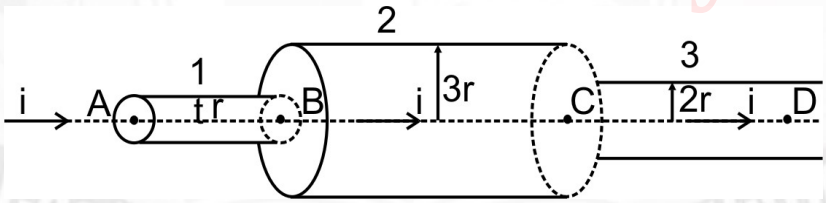
	List – I		List - II
(I)	Value of E_1	(P)	150
(II)	Value of E_2	(Q)	75
(III)	Value of E_1+E_2	(R)	225
(IV)	Value of E_1-E_2	(S)	175
		(T)	450
		(U)	300

If $A_1 = 4m^2$, $A_2 = 8m^2$, $d_1 = 6$mm, $d_2 = 12$mm and applied field is 100 N/C. Match List I with List II.

A) I – R , II – Q, III – U, IV – T **B)** I – P, II – Q, III – R, IV – Q

C) I – P, II – Q, III – R, IV – T **C)** I – R, II – S, III – U, IV – T

33. Three metallic bars 1, 2, 3 are arranged as shown is figure, with total number of free charge carriers in ratio $N_1 : N_2 : N_3 = 1:3:2$; resistivity ratio $\rho_1 : \rho_2 : \rho_3 = 2:1:3$; lengths in ratio $\ell_1 : \ell_2 : \ell_3 = 2:2:3$ for 1, 2, and 3 bars respectively (radius of cross-section shown in figure), carry current i as shown, then match the columns.



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	List – I		List – II
(I)	If P_1, P_2 and P_3 are power dissipated across AB, BC, and CD respectively then $\frac{P_2 P_3}{P_1^2}$	(P)	$\frac{81}{32}$
(II)	If E_1, E_2 and E_3 are magnitude of Electric – fields across AB, BC and CD respectively then $\frac{E_3^2}{E_1 \cdot E_2}$	(Q)	$\frac{1}{32}$
(III)	If $(\upsilon_d)_1, (\upsilon_d)_2$ and $(\upsilon_d)_3$ are drift speeds in bar AB, BC and CD respectively then $\frac{(\upsilon_d)_2 (\upsilon_d)_3}{(\upsilon_d)_1^2}$	(R)	$\frac{1}{4}$
(IV)	If V_1, V_2, V_3 are potential differences across AB, BC and CD respectively then $\frac{V_2 . V_3}{V_1^2}$	(S)	<1
		(T)	$\frac{5}{32}$

A) $I \to QS; II \to P; III \to QS; IV \to RS$

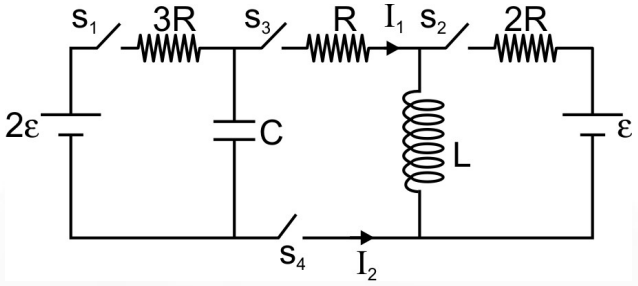
B) $I \to QS; II \to P; III \to RS; IV \to QS$

C) $I \to PS; II \to PS; III \to QS; IV \to RS$

D) $I \to PS; II \to QS; III \to RS; IV \to QT$

34. In the circuit shown S_1 and S_2 are switches which remains closed for a long time. Now S_3 and S_4 are closed simultaneously. Just after S_3 and S_4 are closed, I_1 and I_2 are the currents as shown in circuit. In the List-1 there are some parameters with the answers given in List-II $\left(\varepsilon = 10V, R = \frac{8}{3}\Omega\right)$.

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	List – I		List – II
(I)	I_1 (in amperes) is	(P)	2.5 (in magnitude)
(II)	I_2(in amperes) is	(Q)	50 (in magnitude)
(III)	Rate of change in energy of capacitor is (in J/s)	(R)	Zero
(IV)	Rate of change of magnetic field inside the inductor (in T/s) (assume inductor is a solenoid having 1000 turns per unit length and its volume equal to $\frac{16}{3}\times10^{-3}m^3$	(S)	Positive
		(T)	Negative

A) $I \to PS; II \to PT; III \to QT; IV \to PS$

B) $I \to PS; II \to QT; III \to QT; IV \to PS$

C) $I \to PQ; II \to PT; III \to PS; IV \to RS$

D) $I \to PR; II \to PT; III \to PS; IV \to PT$

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CHEMISTRY **Max Marks: 60**

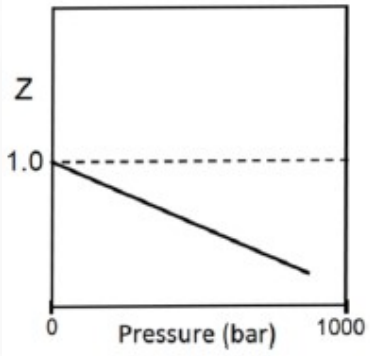
SECTION – II
(SINGLE CORRECT ANSWER TYPE)

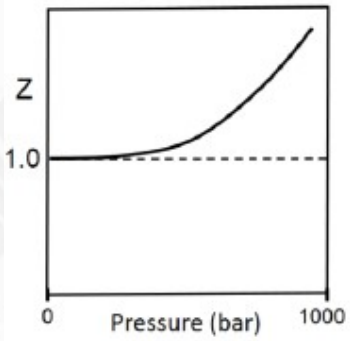
This section contains **FOUR (04)** questions.

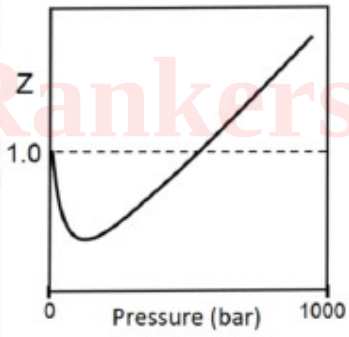
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

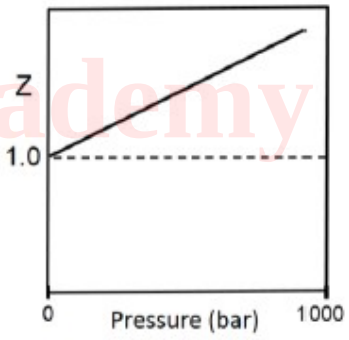
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

35. The Boyle temperature (T_B) is defined as the temperature at which the properties of a real gas coincide with those of an ideal gas in the low pressure limit. The graph that shows the pressure dependence of the compression factor (Z) for a real gas at T_B is

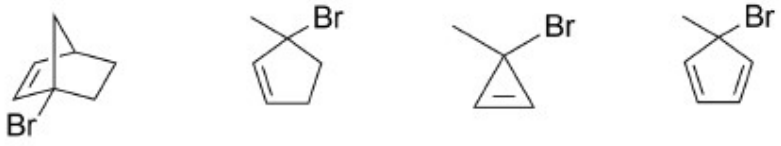
A)


B)


C)


D)


36. The rate of solvolysis of I - IV follows



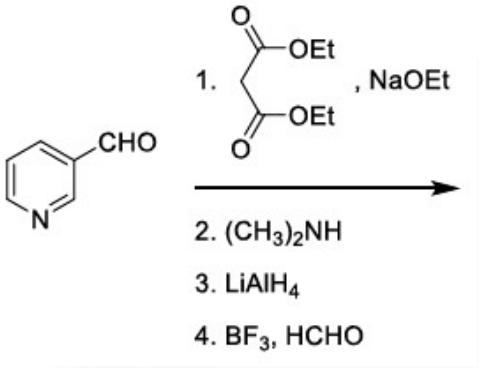
I II III IV

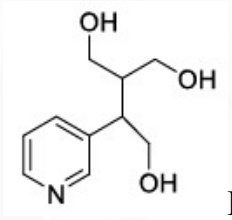
A) I > II > III > IV **B)** III > I > II > IV

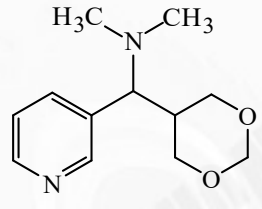
C) III > II > I > IV **D)** IV > I > II > III

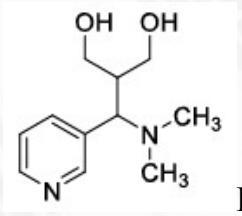
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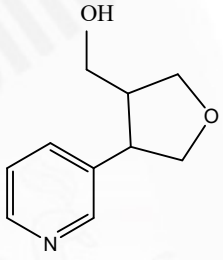
37. The major product formed in the following reaction sequence is



A) 

B) 

C) 

D) 

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

38. AgBr(s) crystallises in rock-salt type structure. At temperature T, a sample of AgBr contains simultaneously both frenkel and schottky defects with the following properties.

2% Ag^+ ions of ideal crystal exhibit frenkel defect and due to frenkel defect all interstitial Ag^+ ions occupy tetrahedral voids.

0.5% of formula unit (AgCl) are completely missing from the crystal lattice due to schottky defect The crystal remains electrically neutral.

Density of perfect AgBr crystal is ρ_0 and density of defective crystal is ρ .

If perfect crystal contains N formula units of AgBr, then how many of the following information's are correct ?

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(i) Total number of Ag^+ ions at normal sites are 0.975 N.

(ii) Total number of Ag^+ at interstitial sites are 0.02 N.

(iii) Total number of Br^- ions present are 0.995 N.

(iv) Value of $\frac{\rho}{\rho^o} = 0.975$.

(v) Fraction of tetrahedral voids occupies by Ag^+ = 0.01 (ignore loss of tetrahedral voids due to defects)

(vi) Coordination number of interstitial Ag^+ ions due to Frenkel defects is 4

A) 3 **B)** 4 **C)** 5 **D)** 6

39. Micelle formation is accompanied by the

A) decrease in overall entropy due to ordering

B) increase in overall entropy due to increase in solvent entropy

C) increase in overall entropy due to increase in solute entropy

D) increase in overall entropy and decrease in enthalpy.

40. Which of the following is (are) dibasic acid(s) with bridging oxo group?

A) pyrosulphuric acid **B)** pyrophosphoric acid

C) pyrophosphorus acid **D)** pyrosulphurous acid

41. How many of the following reaction are not redox reactions?

A) $Pb_3O_4 + conc.HNO_3 \rightarrow$ **B)** $PCl_5 + SO_2 \rightarrow$

C) $HCO_2H \xrightarrow[H_2SO_4]{conc.}$ **D)** $XeF_6 + H_2O \rightarrow$

SECTION-III
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

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42. When the following aldohexose exists in its D-configuration, the total number of stereoisomers in its pyranose form, is



43. Total no. of substance which contain hexagonal planar rings in their structures graphite, $(BN)_x, B_3N_3H_6, C_6H_6, B_2H_6, H_3P_3O_9$:

44. The kinetic energy of a moving electron is 50 kJ. If the de-Broglie wavelength associated with the electron motion is expressed as 0.22×10^{-z} m. What is the value of z?
[Assume that $m_e = 9\times10^{-31} kg$ and h = $6.6\times10^{-34} Js$]

45. 2 mole of a non-volatile solute 'A' are added to 216g of water at 27°C. Vapour pressures of solution are observed as:

Time	0 min	10 min	20 min
Vapour pressure of solution	$6P_0/7$	$4P_0/5$	$3P_0/4$

P_0 = Vapour pressure of pure solvent.

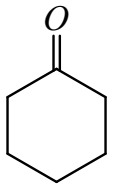
A is dissociated as : $A(aq) \rightarrow 2B(aq) + C(aq)$

'B' and 'C' are also non-volatile and soluble in water. Find rate constant for this reaction?

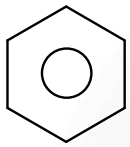
[Unit of rate constant should be mol hr^{-1} or hr^{-1} or $mol^{-1}hr^{-1}$ or $mol^{-2}hr^{-1}$]

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46. Consider the synthesis of ferruginol [take major organic product at each step]



$\xrightarrow[2.CH_3I[3eq]]{1.NaH[3eq]} A$



$\xrightarrow[3.H^+,\Delta]{\begin{array}{l}1.\text{}Cl, AlCl_3\\ 2.O_2, light\end{array}}$ B

$4.NaOH$

$5.CH_3 - I$

$6.CH_3COCl, AlCl_3$

$7.PCl_5$

$8.NaNH_2(2eq)$

[B contains a benzene Ring]

$$B \xrightarrow[2.A\ 3.H_3O^+\ 4.H_2-Pt]{1.NaNH_2} C \xrightarrow[2.alc.KOH]{1.TsCl, py} D \xrightarrow{H^+}$$

$$HI + \underset{ferruginol}{G} \xleftarrow[2.HI]{1.\ \text{  }Cl, AlCl_3} E$$

What is the molar mass of G[ferruginol](g mol^{-1}) Take molar mass : C-12, H-1, O-16

47. Consider the following reaction

$Na_3PO_4 + (NH_4)_2MoO_4 + HNO_3(dil.) \rightarrow 'X'$ (canary yellow ppt.)

Then calculate total number of atoms of 15th group elements which are sp^3 hybridized in compound 'X'.

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases..

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48. Consider products of following reactions.

	COLUMN-I		COLUMN-II
P)	$CoCl_2 + KNO_2 + CH_3COOH \rightarrow$	**1)**	One of product is d^2sp^3 hybridised.
Q)	$Zn + HNO_3(conc.) \rightarrow$	**2)**	One of product is colored gas at room temperature.
R)	$PbS + O_3 \rightarrow$	**3)**	One of product is white precipitate in water.
S)	$NaBH_4 + I_2 \rightarrow$	**4)**	One of product is paramagnetic gas at room temperature.
		5)	None of the product is paramagnetic.

A) $P \rightarrow 1,3; Q \rightarrow 2,4; R \rightarrow 3,4,5,; S \rightarrow 2,4$

B) $P \rightarrow 1,2; Q \rightarrow 2; R \rightarrow 3,5,; S \rightarrow 2,5$

C) $P \rightarrow 1,4; Q \rightarrow 2,4; R \rightarrow 3,4,; S \rightarrow 5$

D) $P \rightarrow 1,5; Q \rightarrow 3,4; R \rightarrow 3,4,; S \rightarrow 5$

49. At 298K, a galvanic cell is constructed as follows:

Pt(s) $|H_2(1atm)|HA(0.1M), NaA(0.1M)||AgCl$ (saturated solution)| Ag(s)

K_a of HA = 10^{-5}, K_{sp} of AgCl= 10^{-10}, K_f of $\left[Ag(NH_3)_2\right]^+ = 10^8$

Volumes of solution in both the compartments are 1L.

$(Use: E^o_{H^+/H_2} = 0, E^o_{Ag^+/Ag} = 0.80V, \frac{2.303RT}{F} = 0.06, \log 3 = 0.50, \log 2 = 0.30$

	List – I		List – II
(P)	Initial EMF of cell	(1)	0.548V
(Q)	Initial $E_{(Anode)}$	(2)	0.80V
(R)	EMF of cell if 0.05 moles of NaOH(s) is added to anodic compartment	(3)	0.30V
(S)	EMF of cell if 2 moles of NH_3 is added to cathodic compartment	(4)	0.83 V
		(5)	0.623V

A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$ **B)** $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$

C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$ **D)** $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

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50.

	COLUMN-I (Reaction)		COLUMN-II (Enthalpy change)
P)	$CO(g)+\frac{1}{2}O_2(g)\rightarrow CO_2(g)$	**1**	$\Delta H_f^o(CO_2)$
Q)	$C(graphite)+O_2(g)\rightarrow CO_2(g)$	**2**	$\Delta H^o_{atomization}(H_2O,l)$
R)	$H_2O(\ell)\rightarrow 2H(g)+O(g)$	**3**	$[\Delta H_f CO_2(g)-\Delta H_f CO(g)]$
S)	$H_2(g)+\frac{1}{2}O_2(g)\rightarrow H_2O(g)$	**4**	$\Delta H_f^o(H_2O,g)$

Codes:

A) P-3; Q-1; R-2; S-4 **B) P-2; Q-3; R-1; S-4**

C) P-1; Q-2; R-3; S-4 **D) P-4; Q-3; R-2; S-1**

51. Match the matrix.

	COLUMN-I		COLUMN-II
A)	$p\pi-p\pi$ bond (s) is/are present	**(P)**	N_2O_3
B)	$p\pi-d\pi$ bond (s) is/are not present	**(Q)**	NO_2
C)	Produce when sulphur react with conc. HNO_3	**(R)**	SO_2
D)	Acidic in nature	**(S)**	N_3H

A) $(A)\rightarrow(PQRS);(B)\rightarrow(PQRS);(C)\rightarrow(PQRS);(D)\rightarrow(PQ)$

B) $(A)\rightarrow(PQRS);(B)\rightarrow(PQS);(C)\rightarrow(QRS);(D)\rightarrow(PQRS)$

C) $(A)\rightarrow(PQRS);(B)\rightarrow(PQS);(C)\rightarrow(Q);(D)\rightarrow(PQRS)$

D) $(A)\rightarrow(PQ);(B)\rightarrow(PQRS);(C)\rightarrow(Q);(D)\rightarrow(PQRS)$

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